

PART-A

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INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3	9	17

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

$$Y_i = \theta_1 x_{i1} + \theta_2 x_{i2} + \epsilon_i$$

$$Y_i = \theta_1 x_{i1} + \theta_2 x_{i2} + \epsilon_i$$

$$X = \begin{bmatrix} 1 & 2 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$\therefore \text{LSE of } \theta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\therefore \hat{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix} = (X^T X)^{-1} X^T Y$$

$$\therefore X^T X = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\therefore \hat{\theta} = (X^T X)^{-1} X^T Y$$

$$= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} y_1 + y_3 \\ y_1 - y_2 \end{bmatrix}$$

$$\therefore \hat{\theta}_1 = \frac{1}{3} (y_1 + y_3) \quad \left. \begin{array}{l} \hat{\theta}_2 = \frac{1}{3} (y_1 - y_2) \end{array} \right\} \underline{Ans}$$

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$2.a) \mathbf{X} = \begin{bmatrix} U_1 \\ U_2 \\ U_1 + U_2 \end{bmatrix} \quad \mathbb{E}(\mathbf{X}) = \begin{bmatrix} \mathbb{E}(U_1) \\ \mathbb{E}(U_2) \\ \mathbb{E}(U_1 + U_2) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 1 \end{bmatrix} \quad \text{Cov}(U_1, U_2) = 0$$

($\because$ Independent)

$$\Sigma = \text{Cov}(\mathbf{X})$$

$$= \mathbb{E}((\mathbf{X} - \mathbb{E}(\mathbf{X}))(\mathbf{X} - \mathbb{E}(\mathbf{X}))^T)$$

$$= \mathbb{E} \left(\begin{bmatrix} U_1 - \frac{1}{2} \\ U_2 - \frac{1}{2} \\ U_1 + U_2 - 1 \end{bmatrix} \begin{bmatrix} U_1 - \frac{1}{2} & U_2 - \frac{1}{2} & U_1 + U_2 - 1 \end{bmatrix} \right)$$

$$= \mathbb{E}$$

$$\text{Var}(U_1) = \mathbb{E}((U_1 - \mathbb{E}(U_1))^2)$$

$$= \mathbb{E}(U_1^2) - (\mathbb{E}(U_1))^2$$

$$\mathbb{E}(U_1^2) = \int_0^1 x^2 dx = \frac{1}{3}$$

$$\text{Var}(U_1) = \frac{1}{3} - \frac{1}{4}$$

$$= \frac{1}{12}$$

$$\Sigma = \text{Cov}(\mathbf{X}) = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

$$\sigma_{11} = \text{Var}(X_1)$$

$$\sigma_{22} = \frac{1}{12}$$

$$= \text{Var}(U_1)$$

$$= \frac{1}{12}$$

$$\sigma_{33} = \text{Var}(U_1 + U_2)$$

$$= \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2)$$

$$= \frac{1}{6}$$

$$\rho_{12} = \frac{\text{Cov}(X_1, X_2)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{22}}} = \frac{\text{Cov}(U_1, U_2)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{22}}} = 0 = \rho_{21}$$

$$\rho_{13} = \frac{\text{Cov}(X_1, X_3)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{33}}} = \frac{\text{Cov}(U_1, U_1 + U_2)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{33}}} = \frac{\text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{33}}} = \frac{\text{Var}(U_1)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{33}}}$$

$$\rho_{23} = \frac{\text{Cov}(X_2, X_3)}{\sqrt{\sigma_{22}} \sqrt{\sigma_{33}}} = \frac{\text{Cov}(U_2, U_1 + U_2)}{\sqrt{\sigma_{22}} \sqrt{\sigma_{33}}} = \frac{\text{Cov}(U_2, U_1) + \text{Cov}(U_2, U_2)}{\sqrt{\sigma_{22}} \sqrt{\sigma_{33}}} = \frac{\sqrt{\sigma_{22}}}{\sqrt{\sigma_{33}}} = \frac{1}{\sqrt{2}}$$

$$= \sqrt{\frac{\sigma_{22}}{\sigma_{33}}} = \frac{1}{\sqrt{2}}$$

$$\therefore \Sigma = \begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} \end{bmatrix}$$

This is actually ρ

As correlation b/w $X_1, X_2 = \text{Corr}(X_1, X_2)$

$$= \frac{\text{Cov}(X_1, X_2)}{\sqrt{\sigma_{11}} \sqrt{\sigma_{22}}} = \frac{\sigma_{12}}{\sigma_{11}} = 1$$

$$\therefore \rho = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \quad \text{Ans}$$

2.b) We need eigenvalues & eigenvectors of $P^T P$

$$P^T P = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{3}{2} & \frac{1}{2} & \sqrt{2} \\ \frac{1}{2} & \frac{3}{2} & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 2 \end{bmatrix}$$

Solve for λ in,

$$\begin{vmatrix} \frac{3}{2} - \lambda & \frac{1}{2} & \sqrt{2} \\ \frac{1}{2} & \frac{3}{2} - \lambda & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 2 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow \left(\frac{3}{2} - \lambda\right) \left[\left(\frac{3}{2} - \lambda\right)(2 - \lambda) - 2 \right] - \frac{1}{2} \left[\frac{1}{2}(2 - \lambda) - 2 \right]$$

$$+ \sqrt{2} \left[\frac{1}{\sqrt{2}} - \sqrt{2} \left(\frac{3}{2} - \lambda\right) \right] = 0$$

$$\Rightarrow \left(\frac{3}{2} - \lambda\right) \left[3 - \frac{3}{2}\lambda - 2\lambda + \lambda^2 - 2 \right] - \frac{1}{2} \left[1 - \frac{1}{2} - 2 \right]$$

$$+ \left[1 - 2\left(\frac{3}{2} - \lambda\right) \right] = 0$$

$$= \left(\frac{3}{2} - \lambda\right) \left[1 - \frac{7}{2}\lambda + \lambda^2 \right] - \frac{1}{2} \left(-1 - \frac{1}{2} \right) + (-2 + 2\lambda) = 0$$

$$\Rightarrow \frac{3}{2} - \frac{21}{4}\lambda + \frac{3}{2}\lambda^2 - \lambda + \frac{7}{2}\lambda^2 - \lambda^3 + \frac{1}{4} + \frac{\lambda}{4} - \frac{1}{2} + 2\lambda = 0$$

$$\Rightarrow -8\lambda + 5\lambda^2 - \lambda^3 = 0$$

$$\Rightarrow \lambda(\lambda^2 - 5\lambda + 8) = 0$$

$$\Rightarrow \lambda(\lambda^2 - 4\lambda - \lambda + 4) = 0$$

$$\Rightarrow -4\lambda + 5\lambda^2 - \lambda^3 = 0$$

$$\Rightarrow \lambda(\lambda^2 - 5\lambda + 4) = 0$$

$$\Rightarrow \lambda^2(\lambda^2 - 4\lambda - \lambda + 4) = 0$$

$$\Rightarrow \lambda^2(\lambda - 4)(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 4, 1, 0$$

Relative

$\therefore$ Variance explained by 1st principal component = $\frac{4}{4+1} = \frac{4}{5}$

$$\text{Total variance} = \frac{1}{2} + \frac{1}{2} + \frac{1}{8} = \frac{1}{3}$$

$$\therefore \text{Variance explained} = \frac{4}{5} \times \frac{1}{3} = \frac{4}{15} \text{ Ans}$$

3. Let the matrix of distance be

$$\begin{array}{c} \begin{matrix} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

3. a.) Here we cluster using max distance

$$\text{Means } d((UV), W) = \max \{ d(UW), d(VW) \}$$

$$\begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{c} \begin{matrix} 12 & 3 & 4 \end{matrix} \\ \begin{matrix} 12 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & \\ 11 & 0 & \\ 5 & 3.5 & 0 \end{bmatrix} \end{array}$$

$$\downarrow$$

$$\begin{array}{c} \begin{matrix} 12 & 34 \end{matrix} \\ \begin{matrix} 12 \\ 34 \end{matrix} \begin{bmatrix} 0 & \\ 11 & 0 \end{bmatrix} \end{array}$$

→ Finally all will be in the same cluster after this step

ii) Here we cluster using avg distance.

$$d((UV), W) = \text{avg}(d(UW), d(VW))$$

$$\begin{array}{c} \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array} \longrightarrow$$

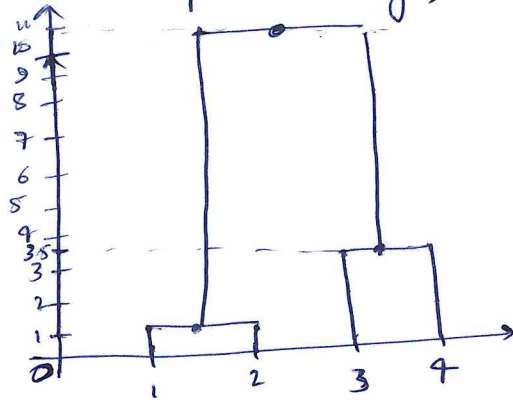
$$\begin{array}{c} \begin{matrix} 12 & 3 & 4 \end{matrix} \\ \begin{matrix} 12 \\ 3 \\ 4 \end{matrix} \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{array} \longrightarrow$$

$$\begin{array}{c} \begin{matrix} 12 & 34 \end{matrix} \\ \begin{matrix} 12 \\ 34 \end{matrix} \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{array}$$

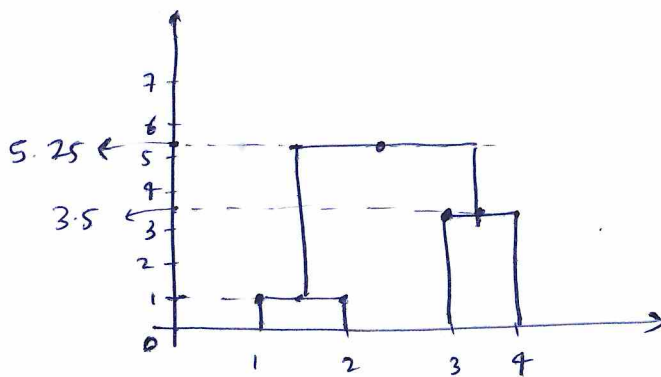
→ Again after this step all clusters will merge into one

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3.b) For complete linkage,



For average linkage,



So as we can see, the clusters formed at each step in both procedures are same but the final cluster in the last step takes lesser distance in average linkage than in complete linkage.

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ADDITIONAL PAGES FOR ANSWERS
